Analysis of Crime, Demographics, and Survey Responses Across U.S. Counties

Prepared for Dr. Hans Oh, Fall 2025

Presenter: Jamie Baldwin & Yvonne Yang

Client: Dr. Hans Oh

Assistant Professor at University of Southern California

- Social worker & psychiatric epidemiologist
- Research Interest:
  - Focuses on social determinants of psychotic experiences
- Education
  - PhD from Columbia University
  - Clinical fellowship training at Yale School of Medicine & VA San Diego
  - Postdoctoral fellowship in prevention science (PIRE / UC Berkeley)

Project Goal: Understanding the Client's Data

Focus on visualization, exploration, and descriptive summaries on different ethnicity

Find extra data information as supplement for the current survey data

Sample Size: 2,801
Data includes: Respondent demographics and responses to the survey (categorical)

Additional Data Sets

Crime Data (2024)

- Across 2,443 Counties
- Include various crime type, such as Violent Crime, Murder, Rape, etc.

Race Data (ACS 5-Year 2023)

- 3,222 Counties
- 8 different race options
- Includes female and male populations
- These are estimates and also include margins of error.

Income Data (ACS 5-Year 2023)

- 3,222 Counties
- Median Household Income based on the Head of the Household's race
- Same race options as the race data. These are also estimates with margins of error.

Analysis Approach

ArcGIS Map

Build an interactive map for the client to help them visualize possible patterns between their survey data and county demographics.

R Shiny

Build an interactive webpage for client to further explore different aspects of the data

Map Pre-process

1. **Find Data**

   Find crime, demographic, and boundary data for counties

2. **Import Data**

   Import all of the datafiles into ArcGIS Pro

3. **Clean Data**

   Use built in processing tools to clean and merge the data

4. **Project Data onto Map**

   Use built in tools to read location data to project the data onto the map

5. **Create Layers**

   Create multiple layers to display different county characteristics

6. **Finalize Map**

   Create an ESRI App to cleanly display the interactive map

Take a look at
the map
together!

R Shiny

1. **Steal Jamie's hard work on data**
2. **Feature Engineering**
   a. Re-code race into three levels: Asian, Black, Two or more race
   b. Calculate the total number of participants in each county
   c. Calculate the percentage of median household income for each ethnicity
   d. Calculate the percentage of each demographic variable of interests corresponds to three different outcomes
3. **Put all information on Shiny**

SHINY

Challenges With Data

- Client said data is **"cleaned"** by a statistician they hired before they sent to us, but it's not.
- All the additional data files are very messy to clean due to spelling errors or inaccurate data entry
  - Additional FBI Crime Data is a very **rough estimate** and needed **severe cleaning leaving room** for incomplete/inaccurate data which can lead to inaccurate assumptions.
- Client has **a specific way** that they would like us to **recode some variables** but never send us more information on that.

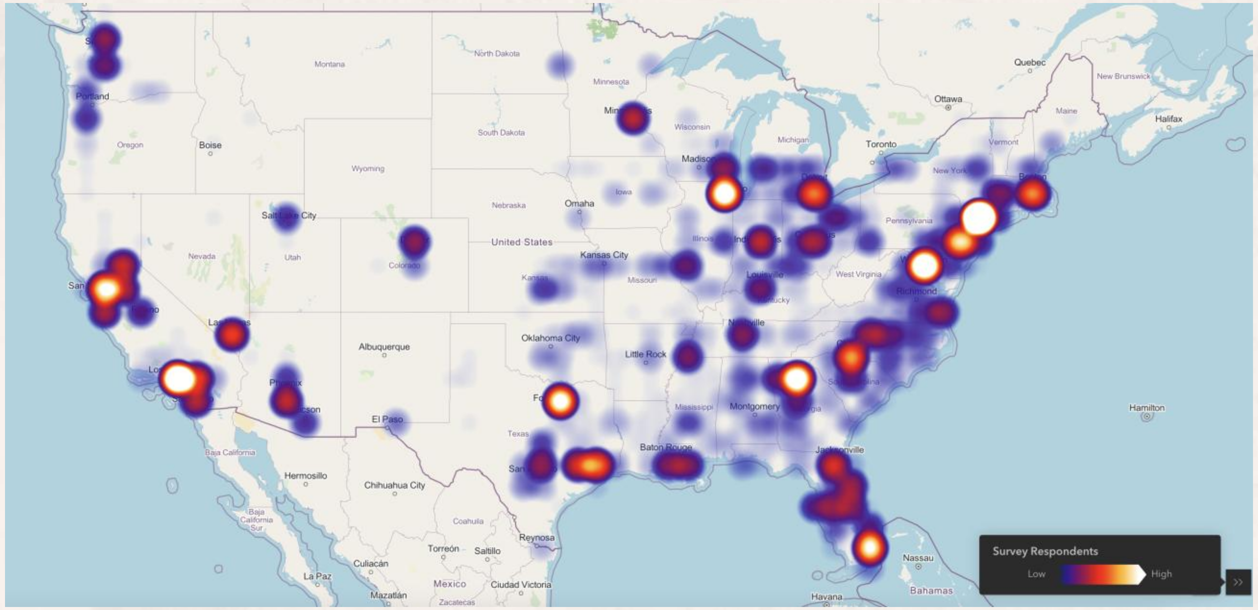
Limitations

- The survey was sent to **random counties**, thus there are many areas to study.
- The majority of the variables are **categorical**, it's hard to be innovative at visualization.
- **Only 5% of the variables** in the dataset are used in this project (the final dataset contains 467 variables)

Final Recommendations

- **Dive Deeper into the Data**
  - They need to get a better understanding of the data and what to do with the data

- **Get more data**
  - Some states like Montana and Wyoming doesn't have any participants.



Potential Future Researches

- **Social Media Track**
  - Gathering more data via social media APIs or through the monitoring of user activity to track the relationship between social media usage and mental health.
- **Start with A Model**
  - The current survey dataset contains detailed information on each participant, including variables like **demographics** and **internet usage.** Those predictors can be used to develop a variety of analytical models.
  - Be cautious of multicollinearity

Thank You